

Pertemuan 12

Auto Scaling

Joko Prasetyo, S.ST., M.Kom.

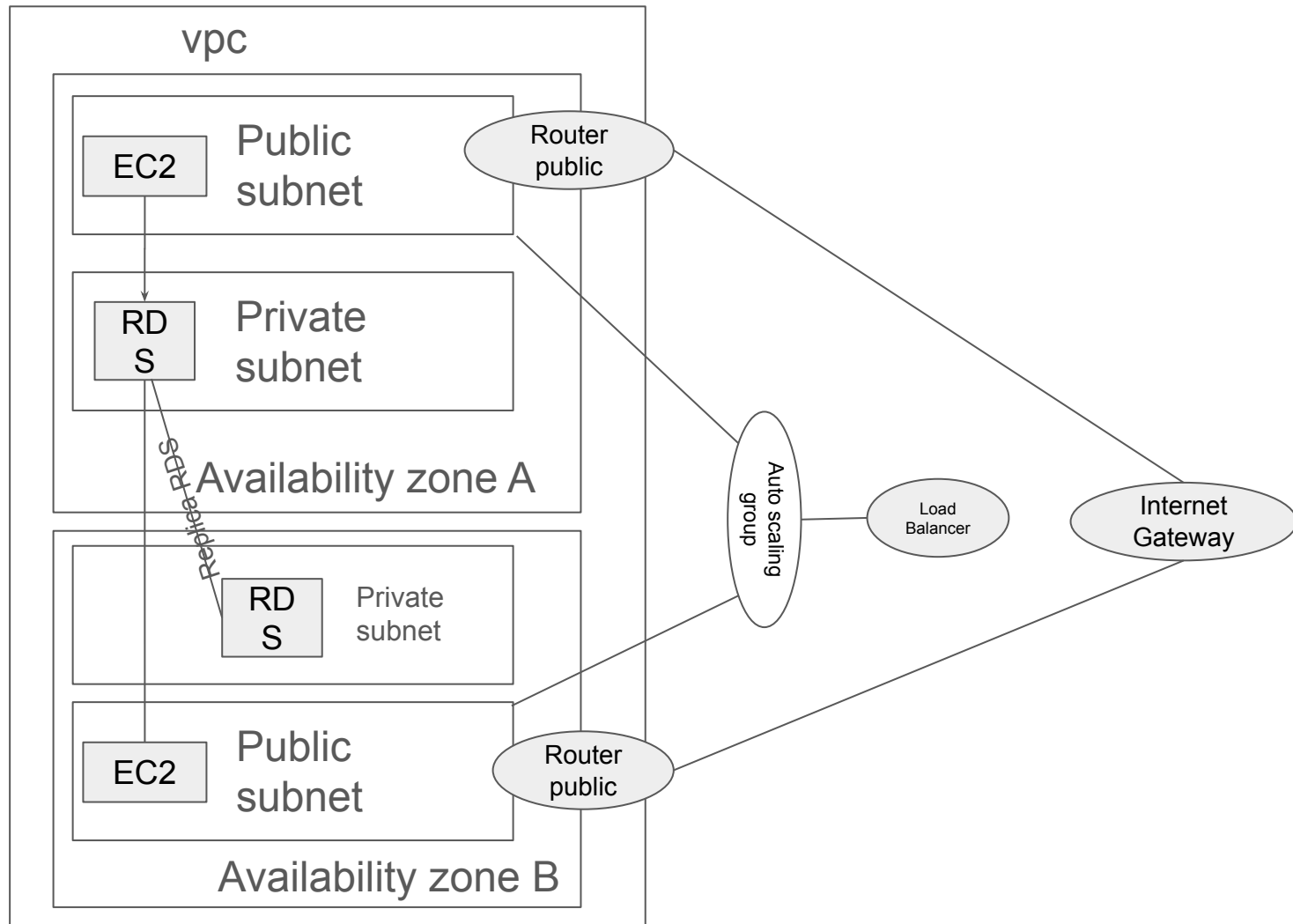
Auto Scaling

Suatu mekanisme yang membuat suatu resource membesar dan mengecil secara otomatis sesuai dengan kondisi-kondisi tertentu

Contoh:

- terdapat 2 server ec2, yaitu ec2 a dan ec2b, yang mana jika resource ec2a dan resource ec2b terpakai semua, maka akan resource ec2 tambahan akan tergenerate otomatis

Arsitektur



provider.tf

provider

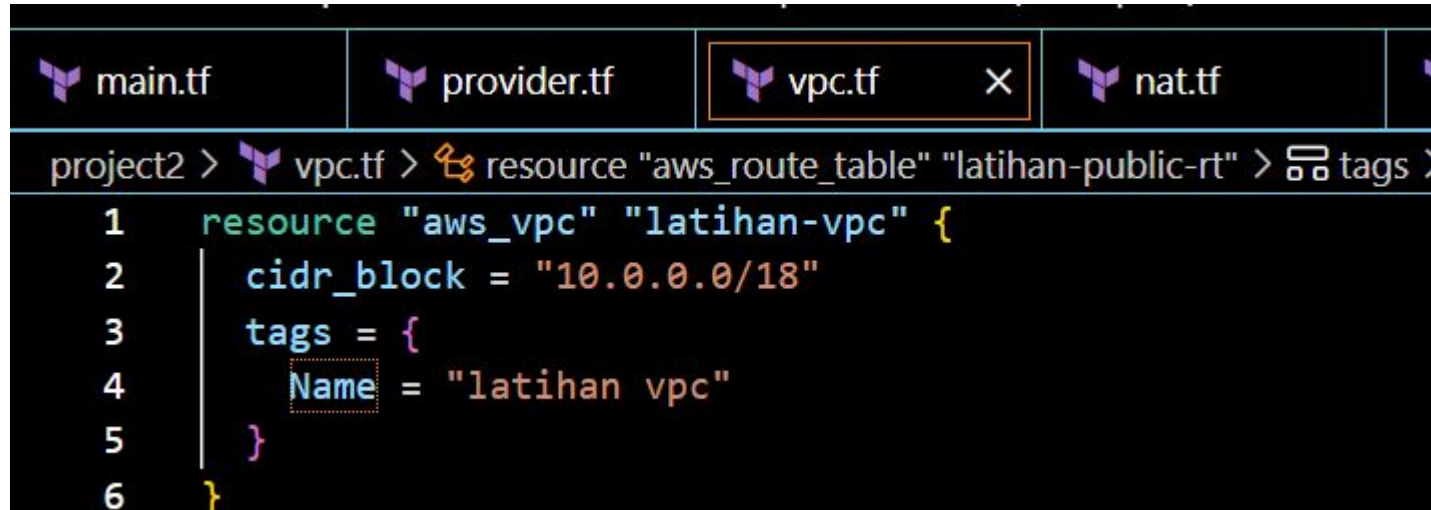
Sesuaikan region

```
1 provider "aws" {  
2     region = "us-west-2"  
3  
4 }  
5  
6
```

vpc.tf

VPC

Ganti latihan-vpc menjadi latihan-vpc-nama_kelompok



```
project2 > vpc.tf > resource "aws_route_table" "latihan-public-rt" > tags >  
1 resource "aws_vpc" "latihan-vpc" {  
2     cidr_block = "10.0.0.0/18"  
3     tags = {  
4         Name = "latihan vpc"  
5     }  
6 }
```

Access control list (ACL)

Mengontrol akses lalu-lintas pada tingkat subnet

Akses inbound:

mysql/mariadb(3306) (allow)

ssh(22) (allow)

Akses outbound:

Semua protokol dan semua port

```
8 resource "aws_network_acl" "latihan-acl" {
9   vpc_id = aws_vpc.latihan-vpc.id
10
11   ingress {
12     protocol = "tcp"
13     rule_no = 100
14     action = "allow"
15     cidr_block = "0.0.0.0/0"
16     from_port = 3306
17     to_port = 3306
18   }
19
20   ingress {
21     protocol = "tcp"
22     rule_no = 101
23     action = "allow"
24     cidr_block = "0.0.0.0/0"
25     from_port = 22
26     to_port = 22
27   }
28
29   egress {
30     protocol = "-1"
31     rule_no = 100
32     action = "allow"
33     cidr_block = "0.0.0.0/0"
34     from_port = 0
35     to_port = 0
36   }
37
38   tags = {
39     Name = "latihan acl"
40   }
41 }
```


Availability Zone a

- Ganti latihan-public-subnet menjadi latihan-public-subnet-nama_kelompok
- Ganti latihan-private-subnet menjadi latihan-private-subnet-nama_kelompok

```
43 resource "aws_subnet" "latihan-public-subnet" {
44     vpc_id = aws_vpc.latihan-vpc.id
45     cidr_block = "10.0.0.0/24"
46     map_public_ip_on_launch = "true"
47     availability_zone = "us-west-2a"
48
49     tags = {
50         Name = "latihan public subnet"
51     }
52 }
53
54 resource "aws_subnet" "latihan-private-subnet" {
55     vpc_id = aws_vpc.latihan-vpc.id
56     cidr_block = "10.0.1.0/24"
57     map_public_ip_on_launch = "false"
58     availability_zone = "us-west-2a"
59
60     tags = {
61         Name = "latihan private subnet"
62     }
63 }
```

Availability Zone b

Ganti latihan-public-subnet2 menjadi
latihan-public-subnet2-nama_kelompok

Ganti latihan-private-subnet2 menjadi
latihan-private-subnet2-nama_kelompok

```
resource "aws_subnet" "latihan-public-subnet2" {  
  vpc_id = aws_vpc.latihan-vpc.id  
  cidr_block = "10.0.2.0/24"  
  map_public_ip_on_launch = "true"  
  availability_zone = "us-west-2b"  
  
  tags = {  
    Name = "latihan public subnet2"  
  }  
}  
  
resource "aws_subnet" "latihan-private-subnet2" {  
  vpc_id = aws_vpc.latihan-vpc.id  
  cidr_block = "10.0.3.0/24"  
  map_public_ip_on_launch = "false"  
  availability_zone = "us-west-2b"  
  
  tags = {  
    Name = "latihan private subnet2"  
  }  
}
```

Asosiasi ACL dengan private subnet

- Menghubungkan acl dan private subnet
- Ganti latihan-acl-assoc menjadi latihan-acl-assoc-nama_kelompok
- Ganti latihan-acl-assoc2 menjadi latihan-acl-assoc2-nama_kelompok

```
76 resource "aws_network_acl_association" "latihan-acl-assoc" {  
77     network_acl_id = aws_network_acl.latihan-acl.id  
78     subnet_id = aws_subnet.latihan-private-subnet.id  
79 }  
80  
81 resource "aws_network_acl_association" "latihan-acl-assoc2" {  
82     network_acl_id = aws_network_acl.latihan-acl.id  
83     subnet_id = aws_subnet.latihan-private-subnet2.id  
84 }  
85
```

Internet gateway

Ganti nama latihan-igw menjadi latihan-igw-nama_kelompok

```
86 resource "aws_internet_gateway" "latihan-igw" {  
87     vpc_id = aws_vpc.latihan-vpc.id  
88  
89     tags = {  
90         Name = "latihan igw"  
91     }  
92 }  
93
```

Router table public

Ganti latihan-public-rt menjadi latihan-rt-nama_kelompok

```
94 resource "aws_route_table" "latihan-public-rt" {
95     vpc_id = aws_vpc.latihan-vpc.id
96
97     route {
98         cidr_block = "0.0.0.0/0"
99         gateway_id = aws_internet_gateway.latihan-igw.id
100     }
101
102     tags = {
103         Name = "latihan public rt"
104     }
105 }
106
```

Asosiasi Public Subnet dan Router Public

Ganti nama latihan-public-rta menjadi latihan-public-rta-nama_kelompok

Ganti nama latihan-public-rta2 menjadi latihan-public-rta2-nama_kelompok

```
120 resource "aws_route_table_association" "latihan-public-rta" {
121     subnet_id = aws_subnet.latihan-public-subnet.id
122     route_table_id = aws_route_table.latihan-public-rt.id
123 }
124
125 resource "aws_route_table_association" "latihan-public-rta2" {}
126     subnet_id = aws_subnet.latihan-public-subnet2.id
127     route_table_id = aws_route_table.latihan-public-rt.id
128 }
129
```

EC2.tf

Security Group untuk EC2

Ganti nama latihan-security-group menjadi latihan-security-group-nama_kelompok

```
1 resource "aws_security_group" "latihan-security-group" {
2   description = "Allow limited inbound external traffic"
3   vpc_id = "${aws_vpc.latihan-vpc.id}"
4
5   name = "latihan_sg"
6
7   ingress {
8     protocol = "tcp"
9     cidr_blocks = ["0.0.0.0/0"]
10    from_port = 22
11    to_port = 22
12  }
13
14  ingress {
15    protocol = "tcp"
16    cidr_blocks = ["0.0.0.0/0"]
17    from_port = 3000
18    to_port = 3000
19  }
20
21  ingress {
22    protocol = "tcp"
23    cidr_blocks = ["0.0.0.0/0"]
24    from_port = 3306
25    to_port = 3306
26  }
27
28  ingress {
29    protocol = "icmp"
30    cidr_blocks = ["0.0.0.0/0"]
31    from_port = -1
32    to_port = -1
33  }
34
35  egress {
36    protocol = "-1"
37    cidr_blocks = ["0.0.0.0/0"]
38    from_port = 0
39    to_port = 0
40  }
41
42  tags = {
43    Name = "latihan_sg"
44  }
45
46 }
```


Key Pair

Ganti nama latihanKeyPair menjadi latihan_key_pair_nama_kelompok

```
48 resource "tls_private_key" "rsa" {
49     algorithm = "RSA"
50     rsa_bits  = 4096
51 }
52
53 resource "local_file" "LatihanPrivateKeyPair" {
54     filename = "latihanKeyPair"
55     content  = tls_private_key.rsa.private_key_pem
56 }
57
58 resource "aws_key_pair" "latihanKeyPair" {
59     key_name = "latihanKeyPair"
60     public_key = tls_private_key.rsa.public_key_openssh
61 }
62
```

Data template file

Data variable yang dapat digunakan untuk memasukkan variable ke suatu file, dalam kasus ini variable dimasukkan ke dalam file [scriptku.sh](#)

Ganti user_data menjadi
user_data_nama_kelompok

```
63 data "template_file" "user_data" {
64     template = "${file("scriptku.sh")}"
65     vars = {
66         rds_address = "${aws_db_instance.latihan_db_rds.address}"
67         rds_username = "${aws_db_instance.latihan_db_rds.username}"
68         rds_password = "${aws_db_instance.latihan_db_rds.password}"
69         rds_db_name = "${aws_db_instance.latihan_db_rds.db_name}"
70     }
71 }
```

scriptku.sh

```
1  #!/bin/bash
2  sudo apt-get update
3  sudo apt install default-mysql-client -y
4  #sudo mysql -u "${rds_username}" --password="${rds_password}" --host "${rds_address}" -e "CREATE DATABASE ${rds_db_name} /*!40100 DEFAULT CHARACTER SET utf8 */;"
5  sudo apt install nodejs npm -y
6  cd /home/admin
7  sudo mkdir myapp
8  cd myapp
9  git clone https://github.com/jokoprsty/latihan_rds.git
10 cd latihan_rds
11 echo "DB_USER=${rds_username}" >> .env
12 echo "DB_PASS=${rds_password}" >> .env
13 echo "DB_NAME=${rds_db_name}" >> .env
14 echo "DB_HOST=${rds_address}" >> .env
15 sudo npm install
16 sudo npm install pm2 -g
17 sudo pm2 start /home/admin/myapp/latihan_rds/app.js -u admin --watch
18 sudo pm2 save
19 pm2 startup
```

EC2 instance (update)

- Meletakkan EC2 instance pada public subnet
- Hapus yang di dalam tanda kotak merah

```
73 locals {
74   loc_ami = "ami-03e383d33727f4804"
75   loc_instance_type = "t2.micro"
76 }
77
78 resource "aws_instance" "latihan-ec2" {
79   depends_on = [ aws_db_instance.latihan_db_rds ]
80   ami = local.loc_ami
81   instance_type = local.loc_instance_type
82   key_name = aws_key_pair.latihanKeyPair.key_name
83   vpc_security_group_ids = [ "${aws_security_group.latihan-security-group.id}" ]
84   subnet_id = "${aws_subnet.latihan-public-subnet.id}"
85   user_data = data.template_file.user_data.rendered
86   tags = {
87     Name = "Latihan-ec2"
88   }
89 }
90
91 resource "aws_instance" "latihan-ec2-2" {
92   depends_on = [ aws_db_instance.latihan_db_rds ]
93   ami = local.loc_ami
94   instance_type = local.loc_instance_type
95   key_name = aws_key_pair.latihanKeyPair.key_name
96   vpc_security_group_ids = [ "${aws_security_group.latihan-security-group.id}" ]
97   subnet_id = "${aws_subnet.latihan-public-subnet2.id}"
98   user_data = data.template_file.user_data.rendered
99   tags = {
100     Name = "Latihan-ec2-2"
101   }
102 }
103
```

rds.tf

Subnet group RDS

- Subnet group berisi lokasi private subnet yang akan digunakan oleh RDS, biasanya lebih dari satu private subnet, dengan masing-masing private subnet berada pada availability zone yang berbeda
- Ganti nama latihan_subnet_db menjadi latihan_subnet_db_nama_kelompok

```
1 resource "aws_db_subnet_group" "latihan_subnet_db" {
2     name           = "latihan_subnet_db"
3     subnet_ids     = [aws_subnet.latihan-private-subnet.id,aws_subnet.latihan-private-subnet2.id]
4
5     tags = {
6         Name = "latihan_subnet_db"
7     }
8 }
```

Security group untuk RDS

Ganti nama rds menjadi
latihan-rds-sg-nama_kelompok

```
10 resource "aws_security_group" "rds" {
11     name     = "latihan_rds_sg"
12     vpc_id   = aws_vpc.latihan-vpc.id
13
14     ingress {
15         from_port     = 3306
16         to_port       = 3306
17         protocol       = "tcp"
18         cidr_blocks    = ["0.0.0.0/0"]
19     }
20
21     egress {
22         from_port     = 3306
23         to_port       = 3306
24         protocol       = "tcp"
25         cidr_blocks    = ["0.0.0.0/0"]
26     }
27
28     tags = {
29         Name = "latihan_rds_sg"
30     }
31 }
```

RDS

- multi_az digunakan untuk membuat replikasi RDS ke availability zone lain
- Ganti nama latihan_db_rds menjadi latihan_db_rds_nama_kelompok

```
resource "aws_db_instance" "latihan_db_rds" {
  identifier          = "latihan-db-rds"
  instance_class      = "db.t3.micro"
  allocated_storage   = 20
  engine              = "mariadb"
  engine_version      = "10.6.14"
  username            = "latihan"
  password            = 12345678
  db_name             = "my_project"
  db_subnet_group_name = aws_db_subnet_group.latihan_subnet_db.name
  vpc_security_group_ids = [aws_security_group.rds.id]
  #parameter_group_name = aws_db_parameter_group.latihan_db_param_group.name
  publicly_accessible = false
  skip_final_snapshot = true
  multi_az            = true
}
```


lb.tf

Load balancer

Digunakan untuk menyeimbangkan lalu lintas data ke beberapa server secara merata

Jenis load balancer

```
resource "aws_lb" "latihan-lb" {  
  name           = "latihan-lb"  
  internal       = false  
  load_balancer_type = "application"  
  security_groups = [aws_security_group.latihan-security-group.id]  
  subnets       = [aws_subnet.latihan-public-subnet.id,aws_subnet.latihan-public-subnet2.id]  
}
```

Load balancer target group

```
10 resource "aws_lb_target_group" "latihan-lb-target-group" {
11     name      = "latihan-lb-target-group"
12     port      = 3000
13     protocol  = "HTTP"
14     vpc_id    = aws_vpc.latihan-vpc.id
15 }
```

Mengasosiasi target group dengan instance (hapus)

Hapus script berikut

```
17 resource "aws_lb_target_group_attachment" "latihan-lb-tga1" {
18     target_group_arn = aws_lb_target_group.latihan-lb-target-group.arn
19     target_id        = aws_instance.latihan-ec2.id
20     port             = 3000
21 }
22
23
24 resource "aws_lb_target_group_attachment" "latihan-lb-tga2" {
25     target_group_arn = aws_lb_target_group.latihan-lb-target-group.arn
26     target_id        = aws_instance.latihan-ec2-2.id
27     port             = 3000
28 }
29
```

Load balancer listener

```
30 resource "aws_lb_listener" "latihan_lb_listener" {
31     load_balancer_arn = aws_lb.latihan-lb.arn
32     port              = "3000"
33     protocol          = "HTTP"
34
35     default_action {
36         type            = "forward"
37         target_group_arn = aws_lb_target_group.latihan-lb-target-group.arn
38     }
39 }
```

autoscaling.tf

Launch template

Menentukan image/os yang dipake serta konfigurasinya

Ganti latihan-launch-template menjadi latihan-launch-template-nama_kelompok

```
resource "aws_launch_template" "latihan-lauch-template" {  
  depends_on = [ aws_db_instance.latihan_db_rds ]  
  name = "latihan-alt"  
  image_id = local.loc_ami  
  instance_type = local.loc_instance_type  
  key_name = aws_key_pair.latihanKeyPair.key_name  
  
  network_interfaces {  
    associate_public_ip_address = true  
    security_groups = [aws_security_group.latihan-security-group.id]  
  }  
  user_data = base64encode(data.template_file.user_data.rendered)  
}
```


Autoscaling group

Menentukan ukuran dari resource yang akan digenerate

Ganti latihan-autoscaling-group menjadi
latihan-autoscaling-group-nama_kelompok

```
resource "aws_autoscaling_group" "latihan-autoscaling-group" {  
  name = "latihan-autoscaling-group"  
  max_size = 10  
  min_size = 2  
  desired_capacity = 2  
  health_check_grace_period = 300  
  health_check_type = "EC2"  
  vpc_zone_identifier = [aws_subnet.latihan-public-subnet.id,aws_subnet.latihan-public-subnet2.id]  
  target_group_arns = [aws_lb_target_group.latihan-lb-target-group.arn]  
  
  launch_template {  
    id = aws_launch_template.latihan-lauch-template.id  
    version = aws_launch_template.latihan-lauch-template.latest_version  
  }  
}
```

Autoscaling policy

- Menentukan aturan kapan resource harus mengecil atau membesar
- Auto scaling berdasarkan penggunaan CPU rata-rata, ada juga pilihan penggunaan traffic jaringan rata-rata, dll, pilih sesuai kebutuhan
- Beri nilai penggunaan cpu rata-rata sebesar 60 persen
- Ganti latihan-auto-policy menjadi latihan-auto-policy-nama_kelompok

```
resource "aws_autoscaling_policy" "latihan-auto-policy" {
  name = "latihan-auto-policy"
  autoscaling_group_name = aws_autoscaling_group.latihan-autoscaling-group.name
  policy_type = "TargetTrackingScaling"

  estimated_instance_warmup = 200

  target_tracking_configuration {
    predefined_metric_specification {
      predefined_metric_type = "ASGAverageCPUUtilization"
    }

    target_value = "60"
  }
}
```

Eksekusi

- terraform init
- terraform plan
- terraform apply (jika terjadi error access denied, lakukan kembali terraform plan)

Jalankan aplikasi web

- Buka browser, kemudian copy alamat dns pada menu load balancer di halaman ec2 ke browser, buka dengan port 3000(jika tidak muncul, tunggu beberapa saat, cek ec2 sampai selesai initializing):
 - `dns_load_balancer:3000/books`
- Insert data ke dalam mysql melalui ec2(gunakan ssh), insert data dengan menggunakan alamat mysql yang ada di `/home/admin/myapp/latihan_rds/.env`, kemudian buka di browser
 - `dns_load_balancer:3000/books/1`
- stop ec2 pada availability zone a, tunggu sebentar sampai status stopped, kemudian cek lagi di browser ke-2 langkah di atas

Cek failover pada RDS

- Lakukan reboot (centang reboot failover) pada RDS instance, kemudian cek pada secondary zone, harusnya berubah ke zona yang lain (agak lama, lakukan refresh berkala)
- Cek kembali di browser langkah pada slide sebelumnya

Aurora and RDS > Databases > latihan-db-rds

DB identifier
latihan-db-rds

Status
Available

Role
Instance

Engine
MySQL

CPU
10.74%

Class
db.t3.micro

Current activity
0 Connections

Region & AZ
us-west-2a

Connectivity & security | **Monitoring** | **Logs & events** | **Configuration** | **Maintenance & backups** | **Data migrations - new** | **Ta**

Instance

Configuration

DB instance ID
latihan-db-rds

Engine version
10.6.14

DB name
my_project

License model
General Public License

Option groups
default:mysql5-7 In sync

Amazon Resource Name (ARN)
arn:aws:rds:us-west-2:692859906669:db:latihan-d
b-rds

Resource ID
db-2GXAJHR3EC5M7A37OM2WTVFP5Q

Created time
May 14, 2025, 20:35 (UTC+07:00)

DB instance parameter group

Instance class

Instance class
db.t3.micro

vCPU
2

RAM
1 GB

Availability

Master username
latihan

Master password

IAM DB authentication
Not enabled

Multi-AZ
Yes

Secondary Zone
us-west-2b

Storage

Encryption
Not enabled

Storage type
General Purpose SSD (gp2)

Storage
20 GiB

Provisioned IOPS
-

Storage throughput
-

Storage autoscaling
Disabled

Storage file system configuration
Current

Install stress test

- Login ke salah satu ec2
- Install stress package:
 - `sudo apt update`
 - `sudo apt upgrade`
 - `sudo apt install stress`
- Jalankan stress test:
 - `sudo stress --cpu 2 --timeout 600`
- mengecek penggunaan cpu, buka terminal baru
 - `top`
- Cek apakah jumlah instance ec2 bertambah di halaman aws, jika bertambah, artinya auto scaling berhasil, karena salah satu ec2 menjalankan resource yang tinggi karena stress test

Cek error

Masuk ssh ke ec2 menggunakan putty, setelah masuk, lihat log dengan mengetikkan menggunakan command:

```
cat /var/log/cloud-init-output.log
```

destroy instance

Perintah jika hanya ingin mendestroy salah satu instance ec2

```
terraform destroy -target aws_instance.<NAME>
```


Tugas

- Buat laporan dari hasil percobaan
- Screenshot pada AWS:
 - VPC
 - Subnet
 - Route
 - Router Association
 - Load balancers
 - Auto scaling group
 - EC2
 - Instance
 - Security group
 - RDS
 - Database
 - Subnet group
 - Security group
- Setelah semua praktikum selesai, lakukan:
 - terraform destroy
- Kumpulkan di https://drive.google.com/drive/folders/1Au7eKYfen94mw_U-btzDvey22WRcKXmf?usp=sharing